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Adoptive Transfer of Immunomodulatory Macrophages Reduces the Pro-Inflammatory Microenvironment and Increases Bone Formation on Titanium Implants

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Abstract

Macrophages play a central role in orchestrating the inflammatory response to implanted biomaterials and are sensitive to changes in the chemical and physical characteristics of the implant. Macrophages respond to biological, chemical, and physical cues by polarizing into pro-inflammatory (M1) or anti-inflammatory (M2) states. We previously showed that rough-hydrophilic titanium (Ti) implants skew macrophage polarization towards an anti-inflammatory phenotype and increase mesenchymal stem cell (MSC) recruitment and bone formation around the implant. In the present study, we aimed to investigate whether the adoptive transfer of macrophages in different polarization states would alter the inflammatory microenvironment and improve biomaterial integration in macrophage-competent and macrophage-ablated mice. We found that ablating macrophages increased the presence of neutrophils, reduced T cells and MSCs, and compromised the healing and biomaterial integration process. These effects could not be rescued with adoptive transfer of naïve or polarized macrophages. Adoptive transfer of M1 macrophages into macrophage-competent mice increased inflammatory cells and inflammatory microenvironment, resulting in decreased bone-to-implant contact. Adoptive transfer of M2 macrophages into macrophage-competent mice reduced the pro-inflammatory environment in the peri-implant tissue and increased bone-to-implant contact. Taken together, our results show the

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Declaration of Competing Interest

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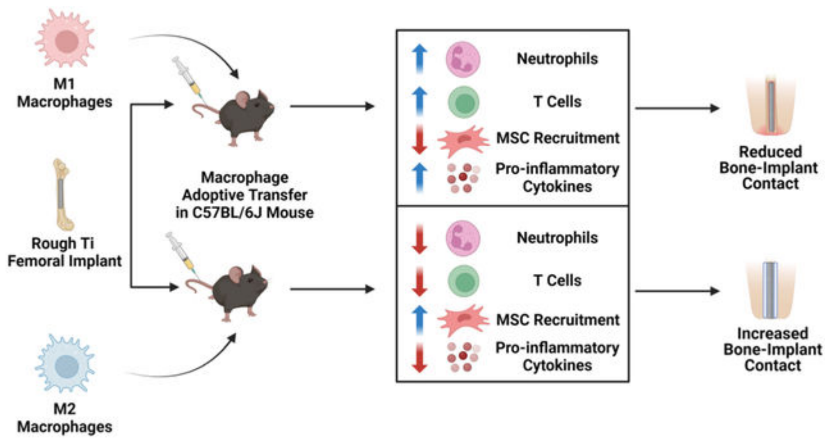
importance of macrophages in controlling and modulating the inflammatory process in response to implanted biomaterials and suggest they can be used to improve outcomes following biomaterial implantation.

Graphical Abstract



Keywords

Macrophages; Titanium; Macrophage ablation; Osseointegration; Adoptive Transfer

1. Introduction

Macrophages are a heterogeneous population of immune cells with remarkable plasticity that play a central role in the inflammatory process in tissue repair following injury, infection, and biomaterial implantation. Macrophages maintain tissue homeostasis by eliminating dead cells and responding to molecular patterns associated with pathogenic infections or cell/tissue damage [1,2]. Macrophages activate or polarize in response to signals in infected or injured tissues, resulting in macrophage phenotypes characterized by differences in surface markers, enzymatic activity, and cytokine release [2–4]. The most-used nomenclature for macrophage polarization, M1/M2, derives from observations that M1 macrophages from prototypical T-helper cell (Th1) strains (C57BL/6J, B10D2) predominantly activate and produce inducible nitric oxide synthase (iNOS), resulting in nitric oxide and Th1 T cell response when exposed to interferon-γ (IFN-γ) or lipopolysaccharides (LPS); while M2 macrophages from Th2 strains (BALB/c, DBA/2) induce arginase to produce ornithine and a Th2 response when exposed to LPS [5].

In vitro studies on macrophage polarization have provided additional mechanistic information. M1 macrophages are activated by exposure to LPS and inflammation-related cytokines TNF-α or IFN-γ, alone or in combination [6,7]. They are characterized by the secretion of toxic effector molecules such as reactive oxygen species, nitric oxide, and inflammatory cytokines (IL-1β, IL-6, IL12, IL-23, TNF-α) [6]. On the other hand, M2 polarization is obtained by exposure to fungal cells, parasites, immune complexes, apoptotic cells, IL-4, IL-13, IL-10, or tumor growth factor beta (TGF-β) [8]. M2 macrophages are characterized by the secretion of anti-inflammatory molecules such as IL-10, TGF-β, and glucocorticoids [9]. However, the M1/M2 paradigm does not reflect the totality of the phenotypic subsets of macrophages as defined by the activating stimuli used and can be further subdivided into M2a, M2b, M2c, M2d, and tumor-associated macrophages [10,11]. Independently of the macrophage M2 subsets, all M2 macrophages produce high levels of IL-10, which is critical in limiting the duration and intensity of immune and inflammatory responses. IL-10 inhibits the production of IL-1β, IL-6, and TNF-α in macrophages by downregulating Major Histocompatibility Complex II (MHC II) and co-stimulatory molecules [10].

Adoptive cell transfer involves isolating immune cells from either the same patient or a donor, which are manipulated and then reinfused into the body. Adoptive cell transfer is commonly used as an immunotherapy for various malignancies [12]. Perhaps the most known application of adoptive cell transfer therapies is genetically engineered T cells expressing chimeric antigen receptors for treating blood cancer [13]. However, macrophages have also been explored for their therapeutic potential since their appearance correlates with the progression of the healing response. Adoptive transfer of M2 macrophages promotes locomotion recovery after spinal cord injury, reduces neuropathic pain, prevents type 1 diabetes, and suppresses experimental autoimmune encephalomyelitis [14–16], demonstrating that M2 macrophages are key regulators in the reparative phases of the healing response. Others have shown the opposing effects of polarized macrophages in a renal injury model, in which M2 macrophages reduced inflammation and injury and M1 macrophages exacerbated kidney injury in an adriamycin-induced kidney disease model [17–21].

Macrophages play a significant role in the inflammatory response following biomaterial implantation, eliminating cellular and tissue debris from the surgical procedure, modulating the inflammatory milieu by regulating cytokine and chemokine release, and recruiting other immune cells. The physicochemical properties of the surface of the implanted biomaterial have been shown to alter macrophage activation and phenotype [22–25]. Our group has demonstrated that macrophages cultured on rough-hydrophilic titanium (Ti) substrates preferentially activate into an M2-like phenotype with higher arginase expression and IL-10 secretion compared to macrophages cultured on hydrophobic Ti substrates, which preferentially activate macrophages into a M1-like phenotype with high secretion of IL-1β, IL-6, and TNF-α, and high expression of nitric oxide synthase (*Nos2*) [24–26]. Importantly, these effects occur *in vitro*, where damage-associated molecular patterns (DAMPs), cytokines, and lipid inflammatory mediators from surrounding tissues and cells are absent [22–24].

We, and others, have shown that macrophages are sensitive to changes in physicochemical properties on the biomaterial surface *in vivo* and that these physicochemical properties can skew macrophage phenotypes [24]. These physicochemical properties not only control macrophage phenotype but also dictate the speed of the healing response and, in some cases, the fate of the implanted material [22,27,28]. In the specific case of Ti implants, we have shown that rough-hydrophilic Ti implants skew macrophage polarization toward an anti-inflammatory or M2-like phenotype and are capable of polarizing $CD4^+$ T cells into

Th2 or regulatory T cells (Treg), cells critical for immunomodulation of the inflammatory process, resulting in faster healing and osseointegration [26,29,30]. We have shown that smooth-hydrophobic Ti implants skew macrophages into a pro-inflammatory M1 phenotype and polarize CD4+ T cells into Th1 or Th17 cells, which extends the inflammatory process and the healing time [26,30].

Taken together, our previous results suggest that the presence of M2-like macrophages at early time points results in a faster resolution of the inflammatory response and implant osseointegration. This study aimed to assess the effect of the adoptive transfer of polarized macrophages on immune cell response and bone formation in a mouse model of Ti implantation.

2. Materials and Methods

2.1 Titanium Implants and Material Characterization

Surface-modified Ti implants (Institut Straumann, Basel, Switzerland) were fabricated from 1 mm unalloyed titanium grade 5 rods as previously described [30–32]. Titanium implants (length=4 mm; Ø=1mm) were degreased in an acetone bath and treated with 2% ammonium fluoride, 2% hydrofluoric acid, and 10% nitric acid to produce smooth Ti implants. Smooth Ti implants were sand-blasted (0.25–0.50 mm coarse-grit corundum) and acid-etched (HCl and H_2SO_4) to produce rough implants. After processing, smooth and rough Ti implants were rinsed in deionized water and dried on the benchtop. To create super-hydrophilic implants, Ti rods were rinsed under a nitrogen atmosphere after sandblasting and acid etching and stored in an isotonic NaCl solution to prevent atmospheric contamination. Implants were sterilized by γ-irradiation before use.

Scanning electron microscopy was used to assess Ti implant surface topography (SEM, Zeiss Auriga, Carl Zeiss, Germany), and confocal microscopy (LSM 910 Laser Scanning Microscope, Germany) to assess surface roughness as previously described [24,26,30,32]. X-ray photoelectron spectroscopy (XPS, PhI5000 VersaProbe spectrometer (ULVAC-PHI, Inc.) was used to assess the oxide layer composition. Data were analyzed using the program CasaXPS. Surface wettability was evaluated by sessile drop contact angle using a Ramé-Hart goniometer (Model 100-25a, Ramé-Hart Instrument Co., Succasunna, NJ).

2.2 Effect of Macrophage Ablation During the Early Inflammatory Response in the Local Inflammatory Cells

12-week-old male C57BL/6J (stock #000664) or Macrophage Fas-Induced Apoptosis (MaFIA) mice (stock #005070, The Jackson Laboratory, Bar Harbor, ME) were injected daily for three days with 5 mg/kg of AP20187 (Cayman Chemical, Ann Arbor, MI) via tail vein injection prior to surgical procedure. Macrophage-competent (C57BL/6J) or macrophage-ablated (MaFIA) mice were anesthetized using isoflurane delivered in O_2 gas and monitored for unconsciousness by pedal reflex. We used macrophage-competent mice to understand normal macrophage activation and macrophage effects on other immune cells and the biomaterial integrative process. The macrophage-ablated model was used to understand the contribution of macrophages in establishing the inflammatory process, recruiting other immune and progenitor cells, and macrophage contribution to the biomaterial integrative process. Surgical dissection of the skin and the patellar tendon was performed to expose the femoral condyles. A dental bur was used to access the medullar canal, and a cylindrical Ti implant (Ø=1mm) with a smooth, rough, or rough hydrophilic surface was then press fit (n=9/implant). Mice were administered 1mg/kg buprenorphine ER LAB (Wedgewood Pharmacy, Swedesboro, NJ) prior to recovery from anesthesia to provide 72 hours of postoperative analgesia. Animals were monitored until initial ambulation and every 24 hours afterward. All animals had access to food and water ad libitum for the duration of the study. No signs of infection were seen in this study. Animal handling procedures were performed under the approval of the Virginia Commonwealth University Institutional Animal Care and Use Committee (Protocol: AD10001108).

Mice were euthanized by CO2 asphyxiation on postoperative day 3 or day 7. Peri-implant tissue cells were isolated by cutting the femur mid-shaft and flushing five times with chilled calcium and magnesium-free PBS (ThermoFisher Scientific, Waltham, MA). Cells were centrifuged at 2000 rpm, and supernatants were aliquoted for inflammatory cytokine assessment. The remaining pellet was resuspended and incubated in Accutase (Innovative Cell Technologies, San Diego, CA) for 30 minutes at room temperature. Implants were incubated in Accutase for 30 minutes to isolate implant-adherent cells. After incubation, implant adherent and peri-implant tissue cells were centrifuged at 2000 rpm, and supernatants were collected and stored at −20°C for cytokine and chemokine analysis. Pelleted cells were resuspended in PBS, and single-cell suspensions were used for flow cytometry.

2.3 Implant-Adherent and Peri-Implant Cell Identification by Flow Cytometry

Fc receptors were blocked in peri-implant tissue single-cell suspensions by incubation with TruStain FcX Plus (BioLegend, San Diego, CA) to prevent non-specific antibody binding. A fixable viability dye was used to exclude dead cells (Zombie NIR, BioLegend). Cells were then incubated with fluorescent antibodies to identify neutrophils ($CD45^+/CD11b^+/Ly6C^+/Ly6G^+$), macrophages ($CD45^+/CD11b^+/MHCII^+/Cd11c^-/F40/80^+$), pro-inflammatory macrophages ($CD45^+/CD11b^+/MHCII^+/Cd11c^-/F40/80^+/CD206^-/CD80^+$), anti-inflammatory macrophages ($CD45^+/CD11b^+/MHCII^+/Cd11c^-/F40/80^+/CD80^-/CD206^+$), T cells ($CD45^+/CD11b^-/CD3^+$), T helper cells ($CD45^+/CD11b^-/CD3^+/CD8^-/CD4^+$), cytotoxic T cells ($CD45^+/CD11b^-/CD3^+/CD4^-/CD8^+$), natural killer T (NKT) cells ($CD45^+/CD11b^-/CD3^+/CD4^-/CD8^-/NK1.1^+$) and mesenchymal stem cells (MSCs, $CD45^-/Sca1^+/CD90^+/CD105^+$) (BioLegend). Samples were analyzed by spectral flow cytometry using a Cytek Aurora CS instrument (Cytek Biosciences, Fremont, CA), and 200,000 events were collected per sample. Single color and fluorescence minus one control were used to establish the gating strategy, as previously reported [30]. Results were analyzed using FlowJo software (FlowJo LLC, Ashland, OR).

2.4. Effect of Macrophage Ablation During the Early Inflammatory Response in the Local Inflammatory Microenvironment

Supernatants obtained from peri-implant tissue were thawed and vortexed, and levels of IL-1β, IL-6, IL-10, TNF-α, CXCL1 (BioLegend), and IL-12p70, CXCL5, CXCL10 (R&D

Systems, Minneapolis, MN) were measured using ELISA kits following manufacturer's instructions.

2.5 Macrophage Isolation

Primary murine macrophages were derived from bone marrow harvested from 12-week-old male C57BL/6J mice (The Jackson Laboratory). Mice were euthanized by CO2 asphyxiation, and femur bones were removed. The bone marrow was flushed with calcium and magnesium-free PBS and treated with ACK lysis buffer (ThermoFisher Scientific). Naïve macrophages were derived by culturing cells in basal media [Dulbecco's Modified Eagle Medium (ThermoFisher Scientific) supplemented with 10% fetal bovine serum (ThermoFisher), 50 U/mL penicillin-50 μg/mL streptomycin (ThermoFisher Scientific) supplemented with 30 ng/mL macrophage colony-stimulating factor (M-CSF, BioLegend) for six days at 37°C, 5% CO2, and 100% humidity. Naïve macrophages were then detached with Accutase and plated for experiments.

2.6 Macrophage Pro- and Anti-Inflammatory Polarization

Two protocols for pro-inflammatory macrophage (M1) activation were performed. Briefly, naïve macrophages were seeded in 24-well plates at a density of 50,000 cells/cm^2 in basal media. Macrophages were then treated for 24 hours with either 20 ng/mL IFN-γ (BioLegend) [33,34] or 20 ng/mL IFN- γ + 100 ng/mL LPS (Sigma-Aldrich, St. Louis, MO) for 24 hours [15,35,36]. Five protocols were used for anti-inflammatory macrophage (M2) activation. Naïve macrophages were seeded in 24-well plates at a density of 50,000 cells/cm^2 in basal media and treated for 24 hours with a) 20 ng/mL IL-4 (BioLegend) [15,36,37]; b) 20 ng/mL IL-10 (BioLegend) [15,33,38]; c) 20 ng/mL IL-4 + 20 ng/mL IL-10 [15,39,40]; d) 20 ng/mL IL-4 + 20 ng/mL IL-10 + 20 ng/mL TGF-β1 (BioLegend) [15,40]; or e) 20 ng/mL IL-4 + 20 ng/mL IL-13 (BioLegend) [15,41,42]. Naïve macrophages were treated with PBS and served as a control. Following treatment, macrophages were washed 3x with warm calcium and magnesium-free PBS, and fresh basal media was added to the wells. After 6 hours, conditioned media was collected for cytokine analysis and macrophages for RNA extraction and gene expression analysis. Confirmation of the macrophage phenotype after the polarizing stimuli was performed by flow cytometry using the previously mentioned phenotypic characterization for pro-inflammatory (CD45$^+$/CD11b$^+$/MHCII$^+$/Cd11c$^-$/F40/80$^+$/CD206$^-$/CD80$^+$), and anti-inflammatory macrophages (CD45$^+$/CD11b$^+$/MHCII$^+$/Cd11c$^-$/F40/80$^+$/CD80$^-$/CD206$^+$).

2.7 Gene Expression of Polarized Macrophages

mRNA was extracted using TRIzol (ThermoFisher Scientific), and 1 μg of RNA was converted to cDNA using an iScript cDNA synthesis kit (Bio-Rad, Hercules, CA). Gene expression of *Il1b, Il6, Tnf, Nos2, Cd38, Il12b, Irf7, Il10, Arg2, Mrc1, Egr2*, and *Chil3* was assessed by qPCR using SsoAdvanced Universal SYBR green supermix (Bio-Rad) and commercially available primers (Bio-Rad PrimePCR™, Supplementary Table 1). Differences were determined by $2^{-\Delta\Delta CT}$ analysis calculated using endogenous housekeeping gene (*B2m*) and naïve macrophages.

2.8 Cytokine Assessment in Conditioned Media from Polarized Macrophages

Conditioned media from polarized and naïve macrophages was first centrifuged at 2000 rpm to remove cell debris. Levels of IL-1β, IL-6, IL-10, TNF-α (BioLegend), IL-12p70, CXCL1, CXCL5, and CXCL10 (R&D Systems, Minneapolis, MN) were measured in the supernatant by ELISA following the manufacturer's instructions.

2.9 Effect of Adoptive Transfer of Naïve and Polarized Macrophages in the Local Inflammatory Response and Bone Formation

Bone marrow cells were isolated from 12-week-old male C57BL/6J mice and differentiated into macrophages, as described in Section 2.5. Macrophages were then cultured for 24 hours in basal media supplemented either pro-inflammatory (20 ng/mL IFN- γ + 100 ng/mL LPS) or anti-inflammatory (20 ng/mL IL-4 + 20 ng/mL IL-10) polarizing cytokines, which were chosen based on the capacity to upregulate IL-1β and TNF-α or upregulate IL-10 and downregulate IL-1β and TNF-α, respectively. Naïve macrophages cultured in basal media for 24 hours served as control cells. After incubation, cells were detached with Accutase and 1×10^6 cells from each treatment were analyzed by flow cytometry confirming pro- ($CD45^+/CD11b^+/MHCII^+/Cd11c^-/F40/80^+/CD80^+/CD206^-$) and anti-inflammatory phenotypes ($CD45^+/CD11b^+/MHCII^+/Cd11c^-/F40/80^+/CD80^-/CD206^+$). The remaining macrophages were labeled with fluorescent CellTracker dye (ThermoFisher Scientific). After cell staining, cells were resuspended at 5×10^6 cells/mL in calcium and magnesium-free PBS and kept on ice until injection.

12-week-old male C57BL/6J and MaFIA mice were injected for three days with 5 mg/kg of AP20187, as described in Section 2.2. Mice were then instrumented with rough Ti implants, as described above. We selected rough implants for these experiments since rough implants are the most used surface modification for orthopaedic and dental implants, increasing the clinical translatability of this work. Labeled naïve, pro-, or anti-inflammatory macrophages (1×10^6 cells/mouse) in 100 μL were adoptively transferred into C57BL/6J or MaFIA mice via tail vein injection (n=8 mice per condition) immediately after surgical procedure. In this experiment, we used macrophage-ablated mice to assess the contribution of M1- and M2-polarized macrophages to the inflammatory microenvironment and effects on other inflammatory cell recruitment and activation and potential effects on bone formation around the implanted biomaterials. The macrophage-competent mouse model was used to assess the impact of activated macrophages in modulating the inflammatory response and bone formation. Mice injected with 100 μL of PBS were used as no-cell control groups.

2.10 Effect of Inflammatory Microenvironment in Polarized Macrophage Phenotype

Mice (n=8/ condition for each strain) were euthanized by CO2 asphyxiation three or seven days post-implantation, and femurs were dissected. Implants were removed, peri-implant tissue flushed, and then incubated in Accutase at room temperature. Single-cell suspensions were obtained, and cells were then incubated with fluorescent antibodies to identify neutrophils, macrophages, pro-inflammatory macrophages, anti-inflammatory macrophages, T cells, T helper cells, cytotoxic T cells, and MSCs, as previously described. Cells were analyzed by flow cytometry.

Additionally, fluorescent-labeled macrophages were sorted and analyzed by gating fluorescent-labeled cells and macrophages characterized as described above. mRNA was extracted from fluorescent-labeled naïve, pro-, and anti-inflammatory macrophages, and 1 μg of RNA was converted to cDNA as described above. Gene expression of *Il1b, Il6, Tnf, Nos2, Tlr2, Tlr4, Tlr9, Ccl2, Csf2, Stat5, Myd88, Nfkb1, Il10, Arg2, Mrc1, Glul, Chil3, Acadm, and Cpt1a* was assessed by qPCR using commercially available primers (Bio-Rad PrimePCR™, Supplementary Table 1). Differences were determined by $2^{-\Delta\Delta CT}$ analysis calculated using endogenous housekeeping gene (*B2m*) and freshly isolated macrophages from naïve animals.

2.11 Effect of Adoptive Transfer of Naïve and Polarized Macrophages on Osseointegration of Ti Implants in Macrophage-Competent and Macrophage-Depleted Mouse Models

On postoperative day 21, mice (n=8/ condition for each strain) were euthanized by CO2 asphyxiation. Femurs were dissected and fixed in 10% formalin (ThermoFisher Scientific) for microCT analysis.

The distal femoral metaphysis of each mouse was scanned at a resolution of 4032 × 2688 pixels (voxel size of 4.00μm) over 180° using a high-resolution aluminum-copper filter, scanning energies of 80kV and 60 μA, 3940 ms exposure time (SkyScan 1276, Bruker, Billerica, MA) as previously reported [30,32,43]. Briefly, images were acquired every 0.5°, and frame averaging was conducted every five frames. Serial two-dimensional cross-sections were assembled into 3D reconstructions and analyzed using SkyScan software and CTAn (Bruker). De-novo bone formation was quantified in a tubular region of interest (ROI) with a length of 3.75mm, an outer diameter of 1.12mm, and an inner diameter of 1.00mm, creating a volumetric sleeve aligned with the centroid of the implant. A threshold of 1000.00–2500.00 Hounsfield units was used to identify bone. 3D representations of bone formation were modeled using Dragonfly (Comet Technologies Canada Inc., Montréal, QC).

2.12 Statistical Analysis

For *in vitro* experiments, six independent replicates per variable (8% variance, 90% power, 15% effect size, α=0.05) were evaluated. For *in vivo* experiments, eight animals per variable (20% variance, 80% power, 30% effect size, α=0.05) will be evaluated. Data are presented as mean ± SD. Statistical analysis was performed using Prism 10 (GraphPad Software, San Diego, CA). Data were first subjected to the Shapiro-Wilk normality test. Results from this test indicate that the data were normally distributed. A one-factor, equal analysis of variance (ANOVA) was used to test the null hypothesis that group means were equal at a significance level of α=0.05, with post-hoc Tukey's HSD test for multiple comparisons.

3. Results

3.1 Titanium Implant Characterization

Increased surface roughness and topography are evident in SEM images on rough and rough-hydrophilic Ti Implants when compared to smooth implants (Figure 1). Assessment of surface average roughness (Sa, Figure 1A) indicated higher roughness for rough Ti implants (Sa= 3.33 μm) and rough-hydrophilic Ti implants (Sa= 3.31 μm) compared to smooth Ti implants (0.6 μm). No qualitative or quantitative differences were observed in roughness between rough and rough-hydrophilic implants. Rough-hydrophilic implants had the highest wettability with a contact angle (θ) of 0° (Figure 1B). Rough Ti implants were the most hydrophobic (θ = 86°), and smooth implants were relatively hydrophobic (θ = 62°). Rough-hydrophilic implants had the lowest carbon content and the highest levels of oxygen and titanium of the implants tested (Figure 1C).

3.2 Macrophage Ablation Alters the Inflammatory Local Environment and Inhibits Immunomodulatory Effect of Rough-Hydrophilic Modification

We first wanted to investigate macrophages' role in the inflammatory process after biomaterial implantation in a macrophage ablation model. We confirmed that macrophages were reduced in MaFIA mice administered AP20187 and found a 92% reduction in macrophage number on the day of surgery, with minimal effect on other innate immune cells such as neutrophils, and macrophage numbers remained reduced through day 5 (Supplementary Figure 1). On day 3 following biomaterial implantation, as expected, macrophages were reduced in MaFIA mice compared to C57BL/6J on all implants. Likewise, pro- and anti-inflammatory macrophages were reduced in MaFIA mice compared to C57BL/6J mice on all implants. In both C57BL/6J mice and MaFIA mice, we observed a decrease in total and pro-inflammatory macrophages and an increase in anti-inflammatory macrophages on rough-hydrophilic implants compared to smooth and rough Ti (Figure 2), but the effect was muted in MaFIA mice.

In C57BL/6J mice, neutrophils were slightly lower on rough-hydrophilic Ti than smooth or rough implants (Figure 2). MaFIA mice had more peri-implant neutrophils than C57BL/6J mice on all Ti implants (Figure 2). C57BL/6J mice showed decreased T cells on rough-hydrophilic implants compared to smooth and rough implants. We observed a significant decrease in T cells in MaFIA mice without differences between the Ti implants tested. From the T cells, we also observed in MaFIA mice an increase in $CD4^+$ T cells and a decrease in $CD8^+$ T cells in all implants tested. In C57BL/6J mice, we observed increased $CD4^+$ and $CD8^+$ T cells on rough-hydrophilic Ti compared to smooth and rough Ti modifications. Ablation of macrophages dramatically altered the effect of Ti surface modifications on neutrophils and T cells, as observed in C57BL/6J mice, where rough-hydrophilic Ti reduced neutrophil presence and increased $CD4^+$ and $CD8^+$ T cells (Figure 2). MSC recruitment was minimal at this time point, and macrophage ablation did not affect the proportion of MSCs at this time point.

We assessed the levels of cytokines in the peri-implant tissue, and we observed a significant decrease in the levels of pro-inflammatory cytokines (IL-1β, IL-6, IL-12p70, IL-17A, TNF-α) and increased IL-10 in peri-implant tissue from C57BL/6J mice instrumented with rough-hydrophilic implants compared to smooth or rough Ti implants (Figure 3). Furthermore, levels of all pro-inflammatory cytokines measured were significantly elevated in MaFIA mice regardless of implant used. We also observed a significant decrease in IL-10 in the peri-implant tissue of MaFIA mice compared to C57BL/6J mice (Figure 3).

On postoperative day 7, we observed a decrease in neutrophils, total and pro-inflammatory macrophages, total T cells, and $CD8^+$ T cells in C57BL/6J mice compared to MaFIA mice. We also observed a significant reduction in those cell phenotypes in mice instrumented with rough-hydrophilic implants (Figure 4). Additionally, we observed an increase of anti-inflammatory macrophages in mice instrumented with smooth and rough Ti and a decrease in mice instrumented with rough-hydrophilic implants. In MaFIA mice, we observed a significant increase in neutrophils, total and pro-inflammatory macrophages, and a decrease in total T cells, $CD4^+$, and $CD8^+$ T cells. Interestingly, while we observed differences in all immune cells measured between surface modifications, especially in rough-hydrophilic Ti, in C57BL/6J mice, we observed a slight increase in anti-inflammatory macrophages and a small decrease in neutrophils in rough-hydrophilic implants compared to smooth and rough Ti in MaFIA mice. Lastly, we observed a significant increase in MSCs in rough and rough-hydrophilic Ti compared to smooth implants in C57BL/6J mice. MSC proportions were the lowest in all implants instrumented in MaFIA mice (Figure 4). In C57BL/6J mice, we observed a decrease in pro- and anti-inflammatory cytokines from smooth to rough, with the lowest levels observed in rough-hydrophilic implants (Figure 5). Interestingly, IL-1β, IL-6, IL-12p70, and TNF-α were significantly elevated in the peri-implant tissue of all implants tested in MaFIA mice compared to the peri-implant tissue of C57BL/6J mice (Figure 5). Levels of IL-17A in MaFIA mice were elevated in rough and rough-hydrophilic implants compared to C57BL/6J. Finally, levels of the anti-inflammatory cytokine IL-10 were reduced in all implants in MaFIA mice compared to C57BL/6J peri-implant tissue (Figure 5).

3.3 Effect of Polarizing Treatments on Macrophage Phenotype and Gene Expression

We assessed the effect of various macrophage polarizing protocols on gene expression related to macrophage phenotype. We observed that INF-γ treatment alone increases the gene expression of genes associated with M1 macrophage polarization (*Il1b, Il6, Tnf, Nos2, Cd38, Il12a, Il12b*, and *Irf7*). However, treatment with INF-γ combined with LPS robustly upregulated the expression of all pro-inflammatory markers (Supplementary Figure 2). Treatment with INF-γ or INF-γ in combination with LPS did not affect anti-inflammatory markers (*Il10, Arg1, Mrc1, Egr2*, and *Chil3*). We explored five M2 protocol treatments and observed that treatment with IL-10 upregulated anti-inflammatory markers the least. On the other hand, the combination of IL-4 and IL-10 upregulated *Arg1, Mrc1, Egr2*, and *Chil3*. Treatments with IL-4 alone or a combination of IL-4 and IL-13 also upregulate anti-inflammatory markers but to a lesser extent than the IL-4 + IL-10 treatment. Treatment of IL-4 + IL-10 + TGF-β1 upregulated all anti-inflammatory markers to a similar extent to IL-4 treatment but also upregulated pro-inflammatory genes such as *Il1b, Il6*, and *Tnf*. Similarly, treatment with IL-4 or IL-10 alone upregulated pro-inflammatory gene upregulation. We also characterized the macrophages exposed to the polarizing stimuli by their expression of CD80 as a pro-inflammatory marker and CD206 as an anti-inflammatory marker. Macrophages treated with interferon-γ + LPS increased the highest expression of CD80 and decreased the expression of CD206 at the cell membrane (Supplementary Figure 3). On the other hand, expression of CD206 at the cell membrane increased after exposure to all alternative activated or anti-inflammatory protocols tested. However, the expression of CD80 was reduced the greatest using IL-4 + IL-10 (Supplementary Figure 3).

3.4 Effect of Polarizing Treatments on Macrophage Cytokine and Chemokine Secretion

Secretion of all pro-inflammatory cytokines was highest in macrophages stimulated with INF-γ and LPS (Supplementary Figure 4). Treatment with INF-γ alone also increased levels of all pro-inflammatory cytokines in conditioned media but not to the same magnitude as treatment with INF-γ and LPS. Treatment with INF-γ and LPS also increased levels of anti-inflammatory IL-10. Treatment with IL-4 and IL-10 increased the highest levels of IL-10 and significantly reduced levels of IL-1β, IL-6, IL-12p70, and TNF-α. Treatments with IL-4 + IL-10 + TGF-β1, IL-4 + IL-13, IL-4, or IL-10 also increased levels of IL-10 and reduced pro-inflammatory cytokines but in a lesser magnitude than IL-4 + IL-10 (Supplementary Figure 4). We also assessed the secretion of chemokines (CXCL1, CXCL5, and CXCL10) in the conditioned media in response to the polarizing treatments. We observed that treatments with INF-γ alone or a combination of INF-γ and LPS increased levels of all chemokines, with the highest levels observed with INF-γ and LPS treatment. Treatment with IL-4, IL-10, IL4 + IL-10, IL-4 + IL-13, and IL4 + IL-10 + TGF-β1 slightly increased levels of all chemokines, all with similar levels and in much less magnitude than treatments with INF-γ alone or a combination of INF-γ and LPS (Supplementary Figure 4).

3.5 Effect of Adoptive Transfer of Naïve and Polarized Macrophages in the Local Immune Cell Composition and Inflammatory Microenvironment

To understand the effect of naïve or polarized macrophages in orchestrating or altering the initial inflammatory response, we used the adoptive transfer of naïve or polarized macrophages into macrophage-competent mice (C57BL/6J) or macrophage-depleted mice (MaFIA) via tail vein immediately after rough Ti instrumentation. Macrophage phenotype was confirmed by flow cytometry after polarizing treatment and before adoptive transfer. At post-surgical day 3, we observed a reduction of neutrophils and pro- and anti-inflammatory macrophages in C57BL/6J mice treated with naïve macrophages compared to PBS control (Figure 6). C57BL/6J mice treated with M1 macrophages showed increased pro-inflammatory macrophages, total T cells, and $CD4^+$ and $CD8^+$ T cells compared to PBS control; adoptive transfer of M2 macrophages reduced neutrophils, pro-inflammatory macrophages, and $CD8^+$ T cells and increased anti-inflammatory macrophages, total T cells, CD4+ T cells, and MSCs compared to PBS control (Figure 6). In macrophage-depleted mice, treatment with naïve macrophages reduced neutrophils, $CD4^+$, and CD8+ T cells and increased total and pro-inflammatory macrophages and total T cells compared to PBS control (Figure 6). Since we observed an increase in CD3+ cells but a decrease of CD4+ and CD8+ at post-surgical day 3, especially after M2 macrophage transfer, we assessed levels of natural killer T (NKT) cells. We observed increased levels of NKT cells in all MaFIA groups, with the highest NKT cells in mice receiving M2 macrophages. Levels of NKT cells were similar in both mouse models and macrophage transfer paradigms on day 7 (Supplementary Figure 5). MaFIA mice treated with polarized M1 macrophages reduced neutrophils and increased total and pro-inflammatory macrophages, T cells, $CD4^+$ and $CD8^+$ T cells, and MSCs compared to PBS control. Treatment with M2 macrophages increased total and anti-inflammatory macrophages, total and $CD8^+$ T cells, and MSCs, and reduced neutrophils and $CD4^+$ T cells compared to PBS control (Figure 6).

We also measured changes in cytokines in response to treatments with macrophages. We observed similar levels of IL-1β, IL-6, IL-10, IL12p70, IL-17A, and TNF-α in C57BL/6J mice treated with naïve macrophages to levels in peri-implant tissue of C57BL/6J mice treated with PBS control (Figure 7). Adoptive transfer of M1 macrophages elevated levels in peri-implant tissue of IL-1β, IL-6, IL12p70, IL-17A, and TNF-α compared to the PBS control group. Treatment with M2 macrophages in C57BL/6J increased IL-10 but led to similar levels of pro-inflammatory cytokines compared to PBS control (Figure 7). Adoptive transfer of naïve macrophages in MaFIA mice showed similar levels of pro- and anti-inflammatory cytokines to the PBS control group. Treatment with M1 macrophages in MaFIA mice increased levels of IL-1β, IL-6, IL12p70, IL-17A, and TNF-α in the peri-implant tissue, while treatment with M2 macrophages slightly reduced IL-1β, IL-6, and TNF-α and increased levels of IL-10 in the peri-implant environment (Figure 7).

At post-surgical day 7, we observed an increase in macrophages, CD4+ and CD8+ T cells, and a decrease in anti-inflammatory macrophages in C57BL/6J mice treated with naïve macrophages compared to PBS control (Figure 8). Treatment with M1 macrophages resulted in increased neutrophils, total and pro-inflammatory macrophages, T cells, CD4+ and CD8+ T cells, and decreased anti-inflammatory macrophages and MSCs compared to PBS control. M2 macrophage treatment decreased neutrophils, total and pro-inflammatory macrophages, and total, CD4+, and CD8+ T cells and increased MSCs compared to PBS control (Figure 8). On the other hand, the adoptive transfer of M1 macrophages in MaFIA mice increased neutrophils, total and pro-inflammatory macrophages, and CD8+ T cells compared to the adoptive transfer of naïve or M2 macrophages. Treatment with M2 macrophages decreased neutrophils, total and pro-inflammatory macrophages, total and CD8+ T cells, and increased MSCs in MaFIA mice compared to the M1 macrophage group (Figure 8). PBS treatment in MaFIA mice highly increased neutrophils and pro-inflammatory macrophages to similar levels as MaFIA mice receiving M1 macrophages. Furthermore, PBS controls showed the least T cells and MSCs in the peri-implant tissue compared to mice administered macrophages. C56BL/6J mice treated with PBS, naïve, or M2 macrophages showed similar levels of pro- and anti-inflammatory cytokines in the peri-implant tissue. On the other hand, C57BL/6J mice treated with M1 macrophages showed elevated levels of pro-inflammatory cytokines and decreased levels of IL-10 compared to PBS control, naïve, and M2 macrophage treatment groups. (Figure 9). Adoptive transfer of naïve macrophages into macrophage-depleted mice reduced IL-1β, IL-6, IL-17A, and TNF-α compared to the PBS control group. Treatment with M1 macrophages reduced levels of IL-1β and increased levels of IL-17A compared to the control group. Finally, treatment with M2 macrophages reduced IL-1β, IL-6, IL12p70, IL-17A, and TNF-α compared to the PBS control group (Figure 9).

3.6 Effect of Post-Implantation Local Inflammatory Environment on Polarized Macrophage Phenotype and Gene Expression

We assessed if exposure to the inflammatory environment as a product of the initial biomaterial implantation affects the phenotype of the transferred naïve, M1, or M2 macrophages. In C57BL/6J mice, naïve macrophages strongly polarized into M1 phenotype at post-surgical days 3 and 7. M2 polarization was also observed in transferred naïve macrophages, but the proportion of M2 polarization was significantly lower compared to M1 polarization (Figure 10A). Transferred M1 macrophages retained their phenotype at post-surgical days 3 and 7, with a minimal transformation into M2 phenotype. On the other hand, we observed a high conversion of transferred M2 macrophages into M1 macrophages on day 3 and an even higher conversion by day 7 (Figure 10A). Naïve macrophages transferred into MaFIA mice strongly polarized into M1 macrophages with minimal conversion into M2 macrophages in both time points measured (Figure 10A). Transferred M1 macrophages in MaFIA mice retain their phenotype the most from all groups tested, with reduced conversion into M2 macrophages. Finally, most M2 macrophages transferred into MaFIA mice converted into M1 macrophages by post-surgical day 3 (Figure 10A).

We assessed the gene expression of transferred naïve, M1, and M2 macrophages after 3 or 7 days of exposure to the initial inflammatory process after Ti implantation. Naïve and M1 macrophages transferred into C57BL/6J mice upregulated genes associated with a pro-inflammatory phenotype (*Il1b, Il6, Tnf, Tlr4, Tl9, Nfkb1*) and downregulated genes associated with an anti-inflammatory phenotype (*Il10, Arg1, Mrc1, Chil3*, Figure 10B). While both naïve and M1 macrophages show upregulation in similar genes, the effect was stronger following the transfer of M1 macrophages. M2 macrophages transferred to C57BL/6J showed the highest upregulation of anti-inflammatory-associated genes at day 3 but also showed upregulation of *Il1b, Tnf, Tlr4*, and *Ccl2* (Figure 10B). The strongest upregulation of pro-inflammatory-associated genes was observed in MaFIA mice following the transfer of M1 macrophages, which also showed the least expression of any of the anti-inflammatory-associated genes measured (Figure 10B). Similarly, naïve and M2 macrophages transferred into MaFIA showed strong upregulation of pro-inflammatory-associated genes and modest upregulation of anti-inflammatory-associated genes such as *Arg1* and *Mrc1*.

3.7 Effect of Adoptive Transfer of Naïve and Polarized Macrophages Titanium Implant Osseointegration

Adoptive transfer of naïve macrophages into C57BL/6J mice resulted in similar levels of bone-implant contact compared to the PBS control group (Figure 11). Treatment with M1 macrophages in C57BL/6J mice resulted in the lowest bone formation around the Ti implant compared to naïve and M2 macrophage treatment and the PBS control group in C57BL/6J mice. In contrast, the adoptive transfer of M2 macrophages resulted in the highest bone-implant contact observed in C57BL/6J mice (Figure 11). In MaFIA mice, the lowest bone formation around the Ti implant was observed in the PBS control group. Adoptive transfer of macrophages increased bone formation around the Ti implant compared to the PBS control group, but there were no significant differences between the macrophage groups (Figure 11).

4. Discussion

In the current work, we have demonstrated that macrophage phenotype plays a fundamental role in regulating the inflammatory microenvironment and altering biomaterial integration. We demonstrate that the adoptive transfer of M2 macrophages into macrophage-competent mice reduces the initial pro-inflammatory microenvironment and infiltration of other immune cells, resulting in a faster resolution of the inflammatory process and higher bone-to-implant contact than mice receiving no cells or naïve macrophages. While the macrophage-ablated model does not represent any human disease or physiological condition, it was used in this study to isolate the contribution of macrophages to the inflammatory resolution and regenerative process following biomaterial implantation. Furthermore, the macrophage-ablated model allowed us to explore the effect of individual macrophage phenotypes on the inflammatory microenvironment in this context.

Adoptively transferred macrophages can infiltrate into inflamed tissues following the chemotactic gradients generated by tissue damage [44,45]. This property makes macrophages attractive candidates for therapy or carriers to attenuate inflammation, and M2 or reparative macrophages to modulate the local immune environment and promote wound healing has been demonstrated in animal and human models [15,46–50]. Our results showed a reduction in pro-inflammatory macrophages and T cells after M2 macrophage adoptive transfer and a decrease in pro-inflammatory cytokines measured in the peri-implant tissue. This effect has been observed previously, where a single transfer of immunosuppressive macrophages protected the development of diabetes in non-obese diabetic (NOD) mice [15]. Results from that study showed that M2 macrophages modulated the inflammatory response in the pancreas by reducing pro-inflammatory macrophages and dendritic cells and T cell proliferation [15]. In another study, adoptively transferred M2 macrophages decreased renal lesions in a model of ischemia-reperfusion injury (IRI) by reducing infiltration of CD4+ and CD8+ T cells and macrophages and downregulating pro-inflammatory cytokines such as IL-1β, IL6, and TNF-α [18]. Furthermore, transplantation of M2 macrophages also promoted tubular epithelial cell proliferation, promoting kidney repair [18,51]. Additionally, we have shown that macrophages orchestrate the inflammatory process and the immunomodulation phase by altering the local microenvironment, chemoattracting other immune cells and MSCs, and polarizing T cells [26,30,52].

We observed an increase in NKT cells in mice receiving M2 macrophages. While the role of NKT cells in biomaterial integration is unclear, it is known that NKT cells and macrophages interact during sterile inflammation, where signals from the microenvironment and CD1d activate NKT cells to produce either inflammatory cytokines (Th1) or immunomodulatory cytokines (Th2) [53]. NKT cells develop in the thymus, and unlike conventional T cells, only a small subset of NKT cells traffic through the lymph nodes under homeostatic conditions, with the largest populations localized in the liver, lungs, spleen, and bone marrow [54]. Mouse NKT cells exhibit robust migration to CXCL12, which is actively produced during the inflammatory phase and most likely after the macrophage ablation in the bone marrow [55,56].

We have shown previously that independent of surface modification in Ti implants, neutrophils and pro-inflammatory macrophages are more prevalent at the early stages of biomaterial implantation, followed by an increase in anti-inflammatory macrophages, Th2, Treg, and reduction of neutrophils and pro-inflammatory macrophages [25,26,30,52]. Interestingly, we do not observe the shift of the pro-inflammatory macrophage prevalent stage into the increased anti-inflammatory macrophages in any of the groups of macrophage-depleted mice in this study. Other studies have reported similar results to our findings, where macrophage depletion results in higher neutrophil infiltration and impaired wound healing [57–64]. Moreover, bone formation around Ti implants was similar between all groups in macrophage-depleted mice but significantly reduced compared to macrophage-competent or PBS control groups, suggesting that a heterogenous population of macrophages is needed to initiate, propagate, and resolve the inflammatory process adequately. A limitation of our study is that our assessment of bone formation omits important parameters such as bone architecture, fibrotic tissue formation, and spatial localization of immune cells in the peri-implant tissue. However, our bone formation analysis provides a 3D assessment of bone formation around the implant and total bone coverage over the implant surface. Additionally, our study focuses on the early stages after biomaterial implantation and omits the possible role of macrophages in long-term implant stability. While the long-term is out of the scope of this work, it is also known that once the wound-healing process is finished, macrophages typically decrease in number and disappear from the wound.

Our results highlight the immunomodulatory capacity of macrophages in homeostasis since the reconstitution of naïve, M1, or M2 macrophages immediately after the surgical procedure resulted in altered immune cell presence and an extended inflammatory microenvironment, which suggests that the inflammatory microenvironment was altered before the surgical procedure during the treatment for macrophage ablation. In the current study, treatment with naïve, M1, or M2 macrophages resulted in significantly higher bone formation around the Ti implant compared to macrophage-ablated mice treated with PBS, suggesting that macrophages, independently of their phenotype, are critical for the healing process and in our case osseointegration of the Ti implant. Our results also showed that treatment with M1 macrophages in macrophage-competent mice increased neutrophil, macrophage, and T cell recruitment, elevating levels of pro-inflammatory cytokines in the peri-implant environment, resulting in the lowest level of osseointegration between the macrophage-competent groups. Therefore, our results suggest that the peri-implant microenvironment is easily skewed into a pro-inflammatory environment that can recruit even more inflammatory cells in what looks like a positive feedback loop. Similar results were obtained in a mouse model of kidney damage where adoptive transfer of M1 macrophages worsened kidney damage [51]. However, ablation of M1 macrophages during the early stages of fracture healing impairs bone repair by altering the cytokine profile in the fracture [60]. In our study, the immunomodulatory and tissue repair effect was mainly observed in macrophage-competent mice compared to macrophage-ablated mice. This suggests that initial pro-inflammatory response and M1 macrophage polarization are required for efficient M2 macrophage immunomodulation and wound healing. In our current study, we demonstrate that short-term ablation of macrophages is sufficient to reduce T cell recruitment, increase neutrophils, and increase pro-inflammatory cytokines in the peri-implant tissue, resulting in decreased levels of immunomodulatory IL-10 compared to macrophage-competent mice receiving a rough-Ti implant. In our study, the macrophage-ablated does not represent any known human disease or physiological condition but represents just a model that allows us to interrogate the effects of adoptively transferred macrophages in the biomaterial integrative process. It is also important to note that while the mouse wound healing process is physiologically similar to human wound healing, there are important differences. Several physiological and environmental factors, including genetics, age, body weight, food intake, and habits, make human wound healing a complex process that is complicated to recapitulate when using young, healthy mice.

We tested different M2 macrophage activating protocols previously reported. Our results revealed a spectrum of M2 macrophage phenotypes with diverse gene expression and pro- and anti-inflammatory molecule secretion. We selected the combination of IL-4 and IL-10 treatment since this combination of cytokines produced the highest IL-10 secretion and significantly reduced pro-inflammatory cytokines and chemokines. Other studies have shown similar results, polarizing macrophages into an M2 phenotype using these cytokines [15,39,40]. The use of IL-4 and IL-10 in macrophage polarization is common, and *in vitro* studies have identified two disparate phenotypes by using either IL-4 (M2a phenotype) or IL-10 (M2c phenotype) [65,66]. Combining IL-4 and IL-10 enhances the expression of immunomodulatory molecules in macrophages [39]. However, other studies have seen a better immunomodulatory effect when IL-4 and IL-10 are combined with TGF-β1 [15]. Exposure of macrophages to IL-4 and IL-13 results in downregulation of common M1 phenotype markers such as IL-1β, TNFα, IL-6, IL-12, and iNOS, and upregulation of arginase-1, IL-10, and mannose receptor [65]. However, studies using macrophages lacking the α subunit of the IL-4 receptor (IL-4rα) showed increased arginase-1 and mannose receptors when cocultured with primary tubular epithelial cells; these IL-4rα$^-$ macrophages failed to upregulate the same molecules when treated with IL-4, suggesting that IL-4 signaling does not up-regulate arginase-1 and mannose receptors. Moreover, IL-4rα–null mice showed a similar degree of macrophage accumulation in the ischemically-injured kidney as wild-type mice, indicating that macrophages exhibit remarkable plasticity in their expression profile depending on the signals at the microenvironment and can achieve an immunomodulatory phenotype in an IL-4-independent fashion [51].

Following the adoptive transfer, macrophages in this study migrated to the injury site, suggesting that DAMPs from peri-implant tissues and neutrophils are the main drivers of macrophage recruitment after biomaterial implantation. The transferred M1 macrophages retained their phenotypic signature in MaFIA and C57BL/6J mice. In contrast, naïve macrophages preferentially acquired an M1 phenotype, and M2 macrophages polarized toward an M1 phenotype. This change of M2 to an M1 phenotype was more pronounced in macrophage-depleted mice. Supporting this, others have shown that the excessive pro-inflammatory microenvironment in injured tissue forced the transferred macrophages to undergo a phenotype switch. The ability of macrophages to adapt to changing cytokine environments has been demonstrated previously [67,68]. This phenomenon has also been observed in *in vitro* polarized macrophages where M1 or M2 macrophages are unable to maintain their phenotype depending on the local environment [51,69]. Bone marrow-derived macrophages are more likely to switch phenotypes due to their proliferative capacity [69]. Cao *et al*. showed that bone marrow-derived M2 macrophages lost their M2 phenotype under proliferative conditions after only one week of culture; in contrast, non-proliferative bone marrow-derived M2 macrophages maintained their M2 phenotype [17]. Further examination using bone marrow-derived M2 macrophages demonstrated that these cells undergo proliferation via M-CSF signaling *in vivo* and lose their immunomodulatory capacity after adoptive transfer in a mouse model of renal injury [69]. Our results demonstrate that in normal physiological conditions, adoptive transfer or anti-inflammatory macrophages may accelerate wound healing and increase bone formation around implanted biomaterials. Future studies using the adoptive transfer of macrophages into mouse models mimicking human conditions with reduced implant integration will provide more information about the translatability of macrophage adoptive transfer.

5. Conclusion

Our results show the importance of macrophages in the immunomodulatory and healing response after biomaterial implantation. Ablation of macrophages compromises the healing and biomaterial integration process and cannot be rescued with adoptive transfer of naïve or polarized macrophages. Finally, the adoptive transfer of M2 macrophages into macrophage-competent mice reduced the pro-inflammatory environment in the peri-implant tissue, reducing the inflammatory process and increasing bone-to-implant contact.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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Statement of Significance

Macrophages are central in orchestrating the inflammatory response to implanted biomaterials and are sensitive to biomaterial chemical and physical characteristics. Our study shows that a deficiency of macrophages results in prolonged inflammation and abolishes bone-biomaterial integration. Adoptive transfer of immunomodulatory macrophages into macrophage-competent mice reduced the inflammatory environment and increased bone-implant contact.

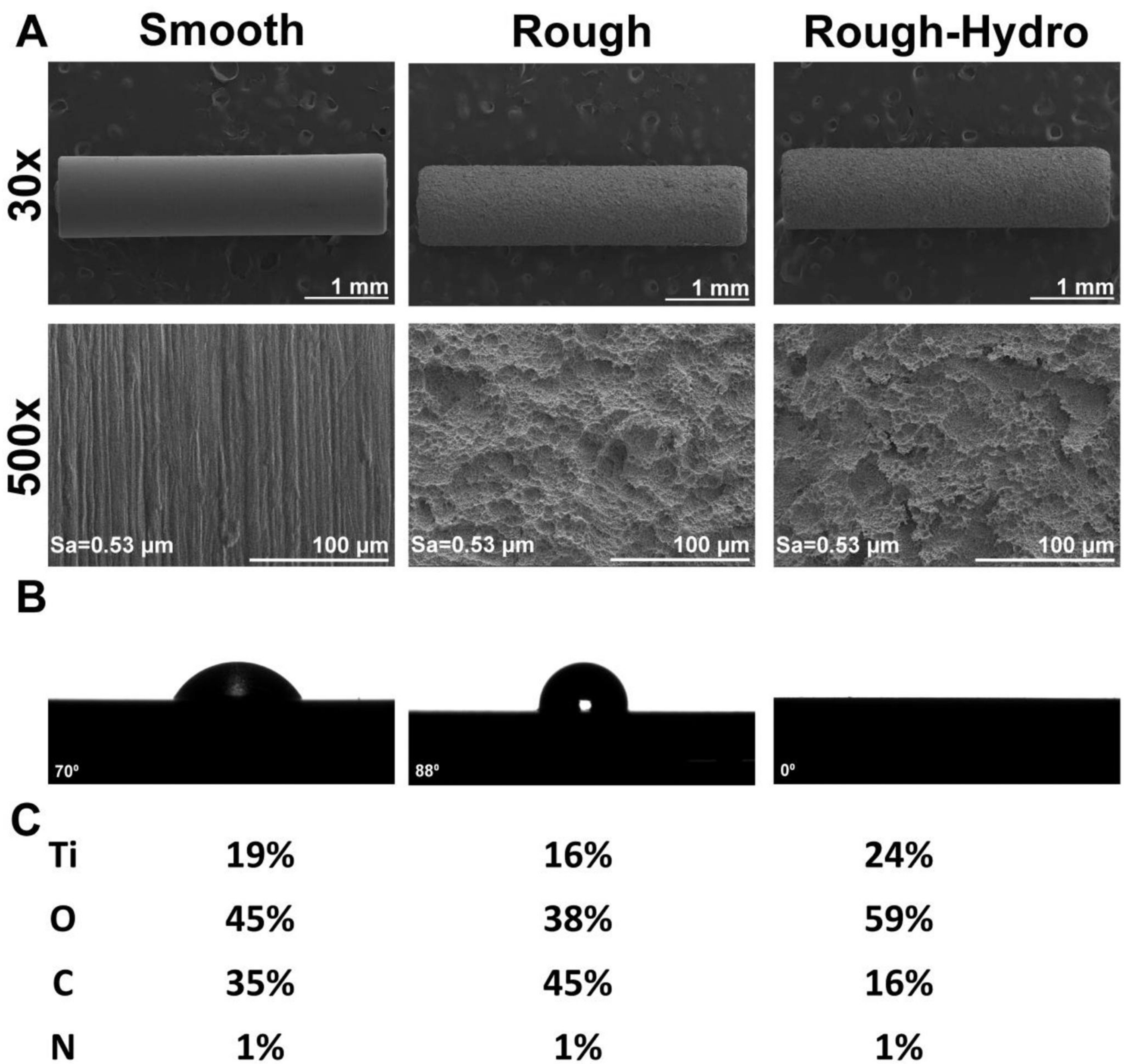

Figure 1:
Surface characterization of smooth, rough, and rough hydrophilic Ti implants. (A) Surface topography was analyzed qualitatively via scanning electron microscopy at 30x and 500x. B) Water contact angle measurement; (C) Elemental chemical composition analysis of the implant oxide layer

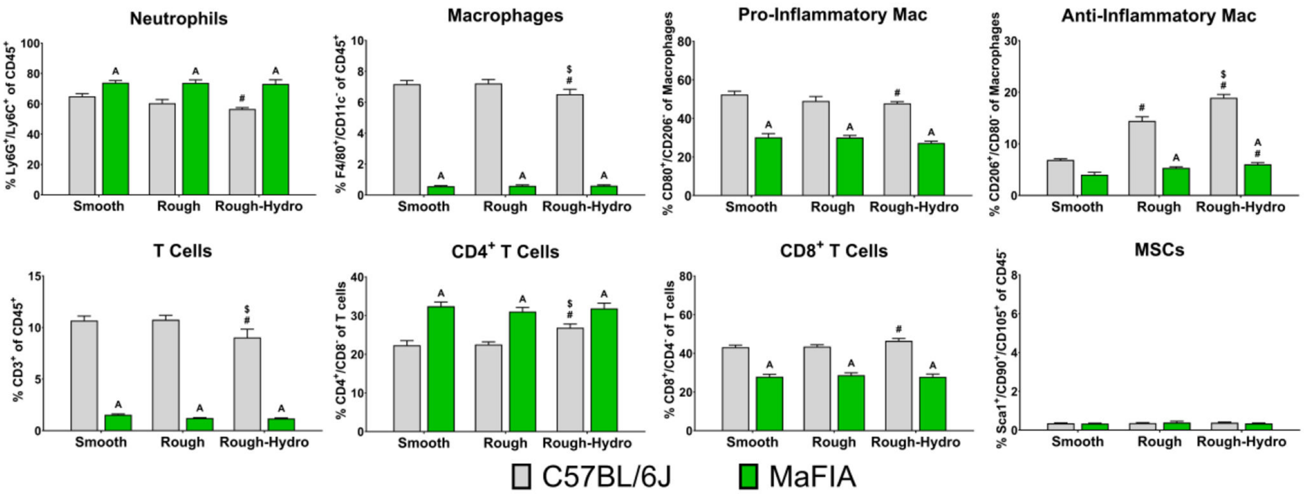

Figure 2:

Macrophage ablation altered peri-implant immune cell population, increasing neutrophils and CD4+ T cells in response to Ti implants. Flow cytometry analysis of peri-implant bone marrow at 3 days post-implantation. $p<0.05$: A vs. C57BL/6J, # vs. smooth Ti, $ vs. rough Ti.

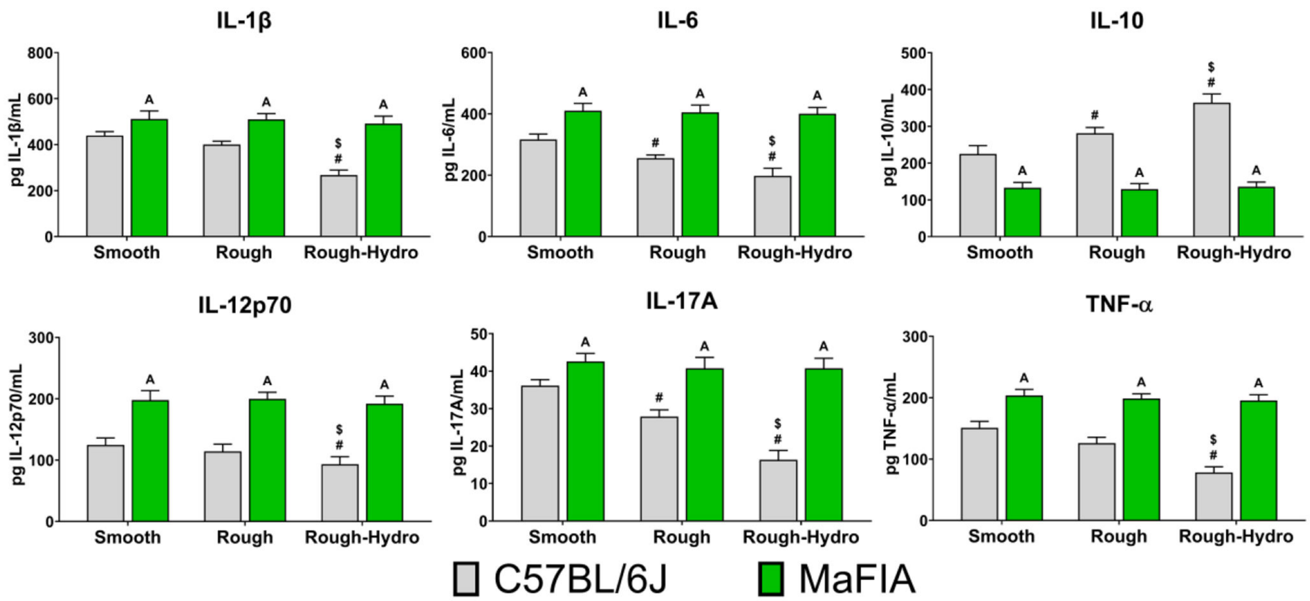

Figure 3:

Inflammatory cytokine production increased in macrophage-ablated mice in response to Ti implants. Cytokine analysis of the local microenvironment was analyzed by ELISA at 3 days post-implantation. $p<0.05$: A vs. C57BL/6J, # vs. smooth Ti, $ vs. rough Ti.

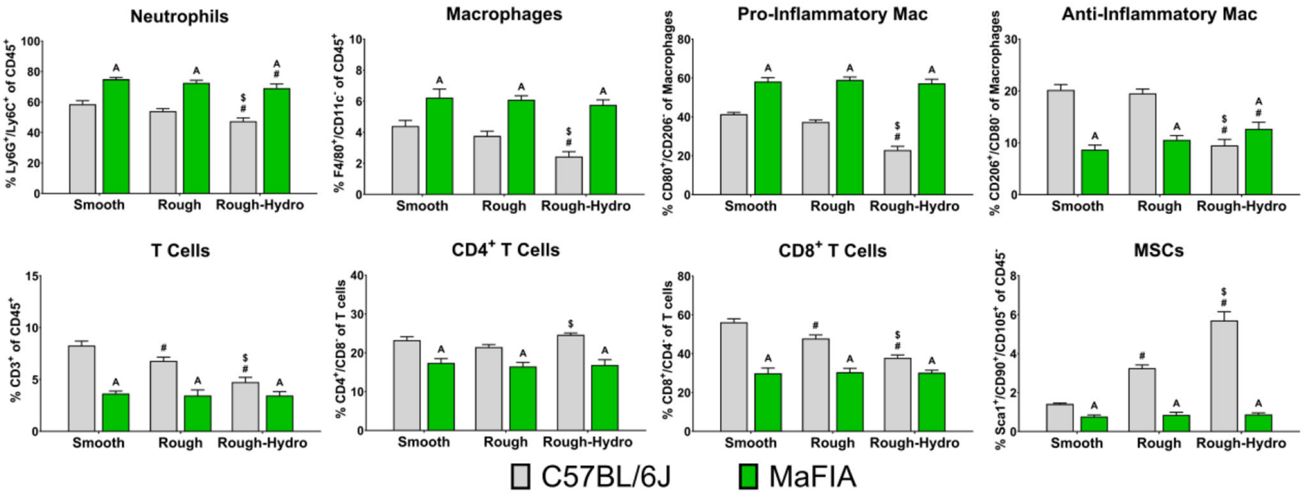

Figure 4:

Macrophage ablation at early inflammatory stages increased the presence of neutrophils and pro-inflammatory macrophages in the peri-implant tissue. Flow cytometry analysis of peri-implant bone marrow at 7 days post-implantation. $p<0.05$: A vs. C57BL/6J, # vs. smooth Ti, $ vs. rough Ti.

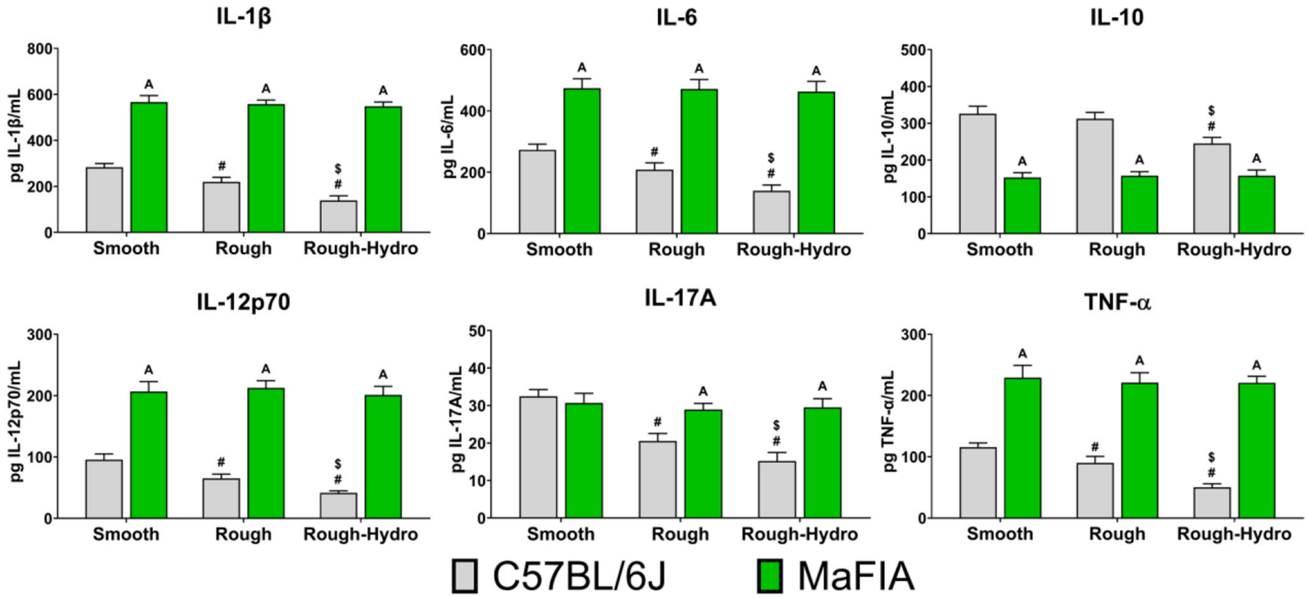

Figure 5:

Macrophage ablation at early inflammatory stages increased levels of inflammatory cytokines in response to Ti implants. Cytokine analysis of the local microenvironment was analyzed by ELISA at 7 days post-implantation. $p<0.05$: A vs. C57BL/6J, # vs. smooth Ti, $ vs. rough Ti.

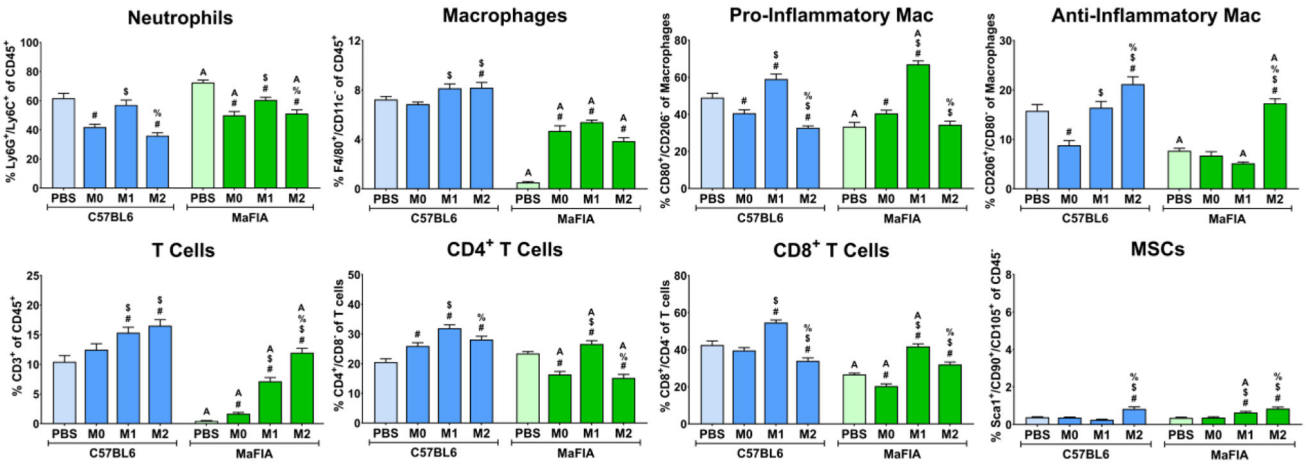

Figure 6:

Adoptive transfer of immunomodulatory macrophages in macrophage-competent mice reduces neutrophil, pro-inflammatory macrophages, and CD8+ T cells in the peri-implant tissue. Flow cytometry analysis of peri-implant bone marrow three days post-implantation and macrophage adoptive transfer. $p<0.05$: # vs. PBS same mouse group, $ vs. M0 same mouse group, % vs. M1 same mouse group, A vs C57BL/6J same treatment.

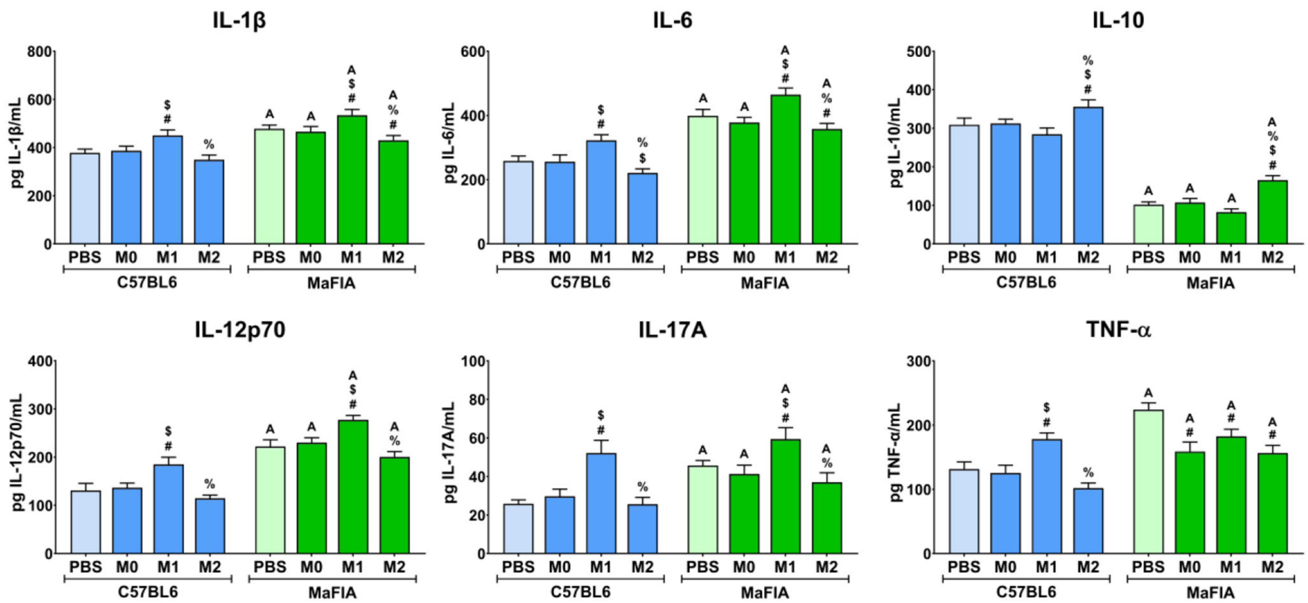

Figure 7:

Adoptive transfer of pro-inflammatory macrophages increases inflammatory cytokines locally. Cytokine analysis of the local microenvironment was analyzed by ELISA at 3 days post-implantation and macrophage adoptive transfer. $p<0.05$: # vs. PBS same mouse group, $ vs. M0 same mouse group, % vs. M1 same mouse group, A vs C57BL/6J same treatment.

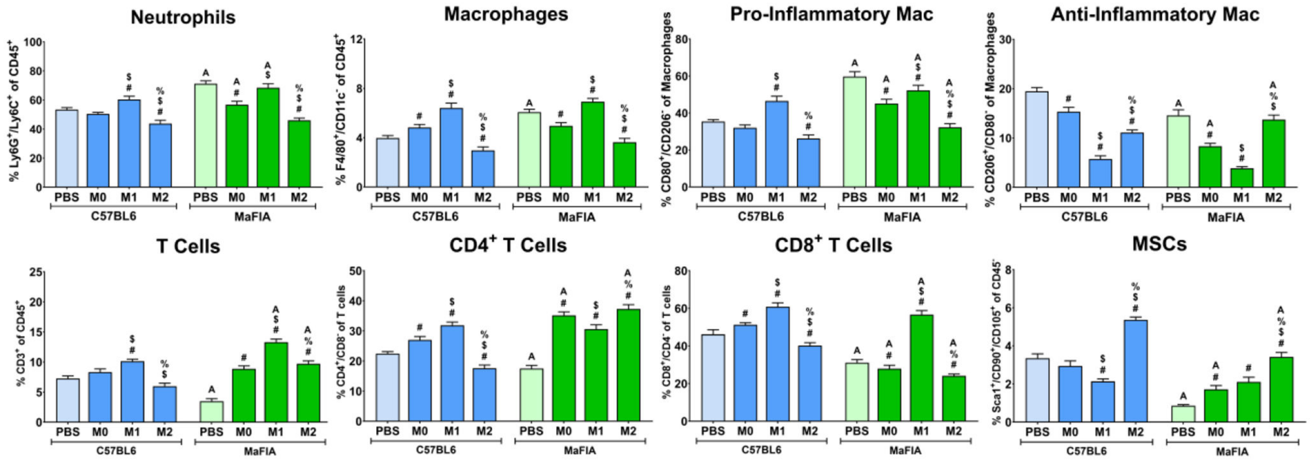

Figure 8:

Adoptive transfer of immunomodulatory macrophages in macrophage-competent mice reduced neutrophils, pro-inflammatory macrophages, and increased MSCs in the peri-implant tissue. Flow cytometry analysis of peri-implant bone marrow at 7 days post-implantation and macrophage adoptive transfer. $p<0.05$: # vs. PBS same mouse group, $ vs. M0 same mouse group, % vs. M1 same mouse group, A vs C57BL/6J same treatment.

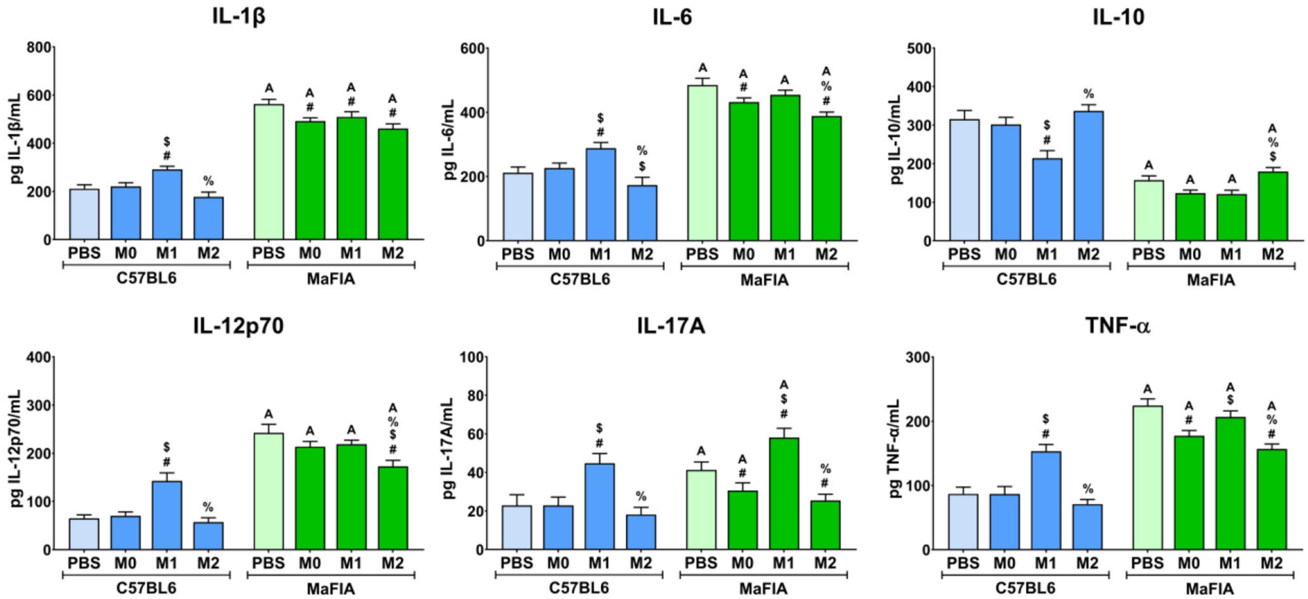

Figure 9:

Pro-inflammatory cytokines in the local microenvironment remain elevated in macrophage-ablated mice at post-surgical day 7. Cytokine analysis of the local microenvironment was analyzed by ELISA at 3 days post-implantation and macrophage adoptive transfer. p<0.05: # vs. PBS same mouse group, $ vs. M0 same mouse group, % vs. M1 same mouse group, A vs C57BL/6J same treatment.

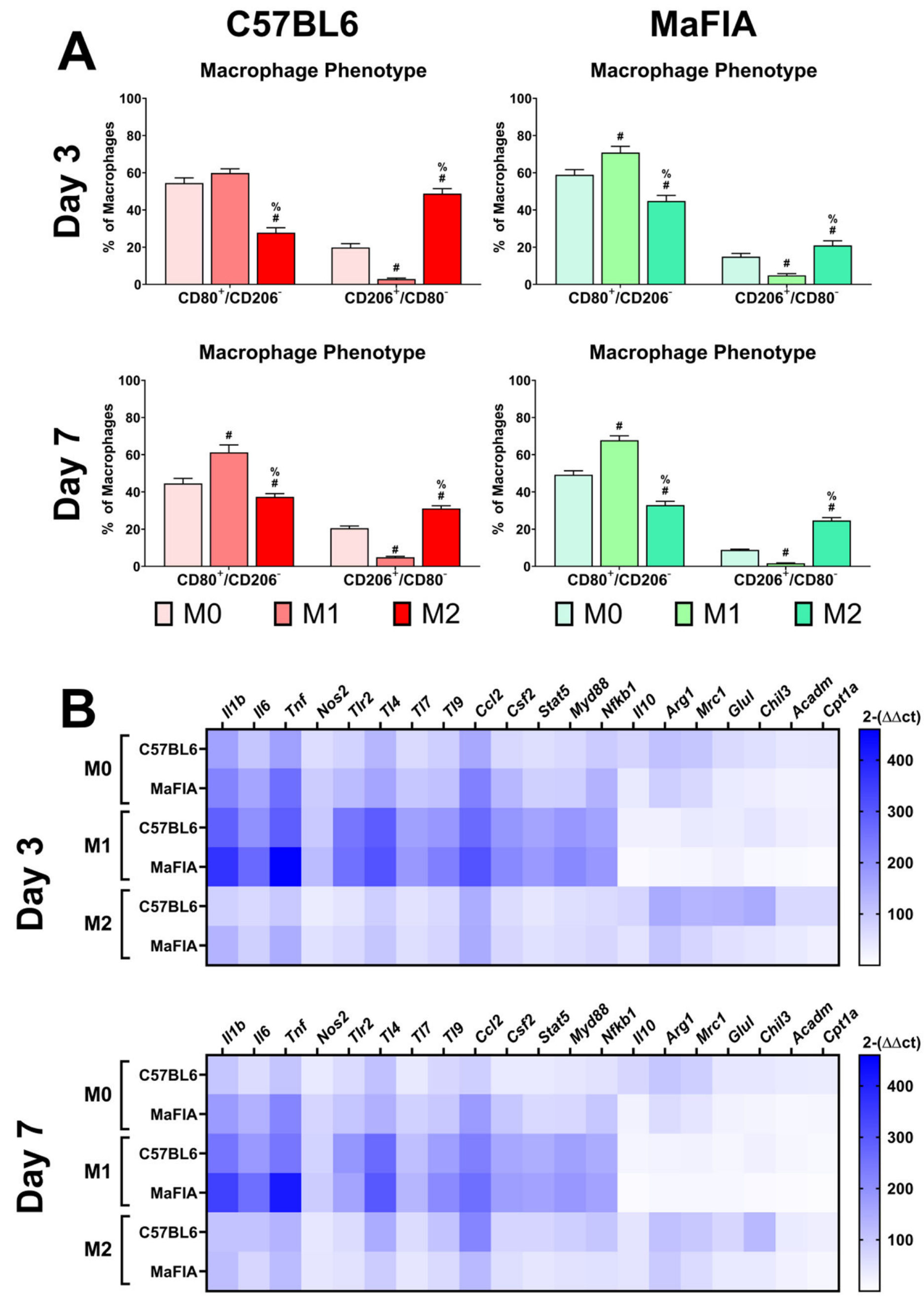

Figure 10:

Adoptive transferred macrophages differentially retain a polarizing phenotype in response to the peri-implant environment. (A) Flow cytometry analysis of labeled-transferred macrophages isolated from the peri-implant tissue at 3 and 7 days post-implantation. (B) Heatmap of gene expression of M0, M1, and M2 transferred macrophages after 3 and 7 days exposed to the peri-implant microenvironment. p<0.05: # vs. M0, $ vs. M1.

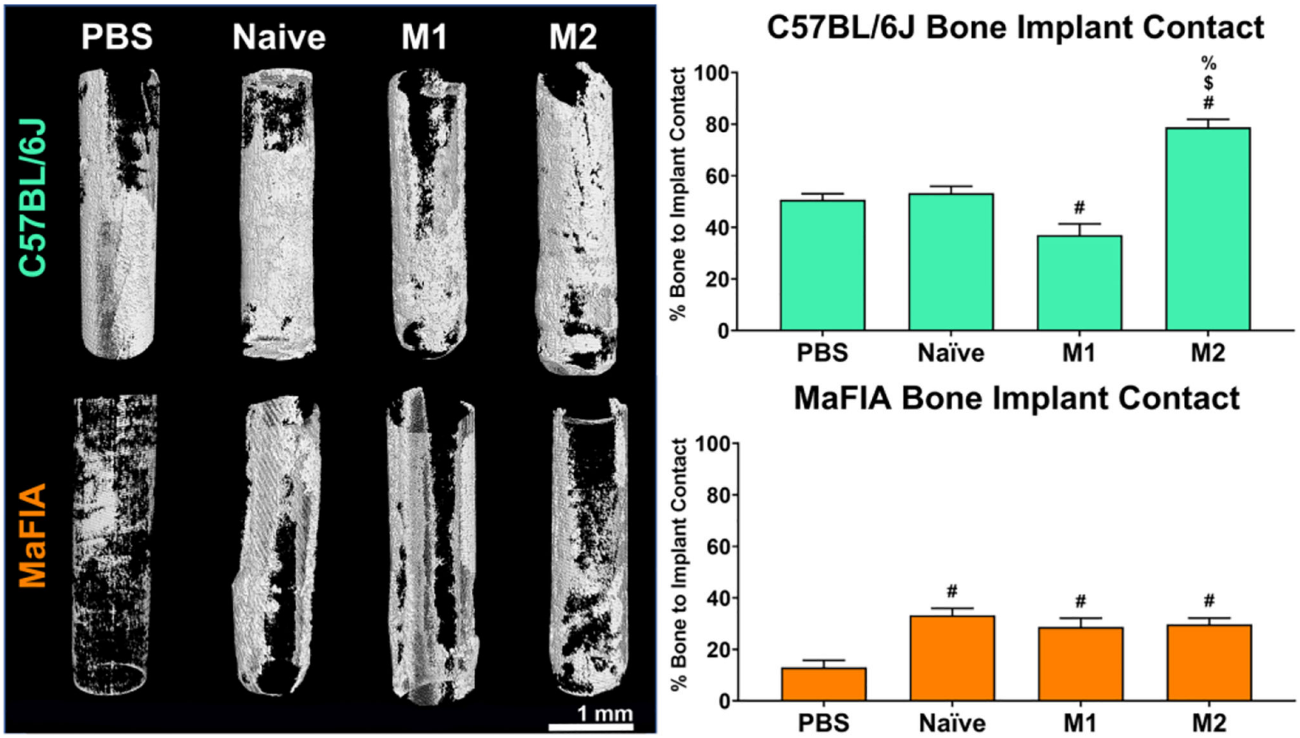

Figure 11:

Adoptive transfer of immunomodulatory macrophages in macrophage-competent mice increases bone-implant contact. Analysis of bone formation around modified Ti implants in macrophage-competent (C57BL/6J) or macrophage-ablated (MaFIA) mice treated with PBS, M0, M1, and M2 polarized macrophages after 21 days post-implantation. $p<0.05$: # vs. PBS, $ vs. M0, % vs. M1.